Healthcare Awareness GenAI Assistant

Assignment Overview

Dear Interns,

As part of your internship evaluation and practical learning, you are required to build a Healthcare Awareness Generative AI Assistant. The assistant should answer general health-awareness questions by using reliable public healthcare information collected and structured by you.

The project should demonstrate your ability to work independently, use open data responsibly, design a usable system, integrate a Generative AI model, and present a demo-ready product.

Project Objective

You have to develop a GenAI-powered healthcare awareness chatbot that can answer general health-related queries using reliable public healthcare information.

• Preventive care and wellness guidance

• Common symptoms and first-level awareness information

• Public health awareness topics

• Government healthcare schemes and public resources

• General, non-diagnostic health education

> **Important:** The bot must not provide medical diagnosis, prescriptions, medicine dosage, or treatment decisions. It should only provide awareness-based information and must recommend consulting a qualified medical professional wherever required.

Data Requirement

The company will not provide any dataset. You are expected to independently collect and use open/public healthcare data from reliable and legally usable sources.

• Government health portals

• Ministry of Health resources

• World Health Organization resources

• Public healthcare PDFs and awareness materials

• Verified health FAQs

• Other legally usable public sources

You must clearly mention all sources used in your final submission and briefly explain how the data was collected, cleaned, structured, and used by the chatbot.

Expected Functionality

• A clean chatbot interface where users can ask healthcare awareness-related questions.

• A backend system to process user queries.

• Integration with a Generative AI model such as OpenAI, Gemini, Claude, Groq, Ollama, or any suitable LLM.

• A knowledge base created from the public healthcare data collected by you.

• Retrieval-based answering, preferably using a RAG approach.

• Source/reference display wherever possible.

• Safe responses for sensitive medical questions.

• A fallback response when the bot does not have enough reliable information.

• Basic chat history for the current session.

• Proper documentation explaining setup, data sources, and system architecture.

Safety and Ethical Guidelines

Since this project relates to healthcare, safety is extremely important. The assistant should avoid overconfident or unsafe medical answers and should make its limitations clear.

• Clearly state that the bot is not a doctor and does not replace professional medical advice.

• For emergency symptoms, serious conditions, medicines, dosage, diagnosis, or treatment-related questions, advise the user to consult a qualified doctor or contact emergency medical services.

• Avoid making definitive claims where the source data is insufficient.

• Prefer cautious, awareness-focused responses with references wherever possible.

• Use safe fallback responses when the query goes beyond the collected data or the bot's intended scope.

Suggested Tech Stack

You may use any technology stack you are comfortable with. The following options are suggested:

Layer	Suggested Options
Frontend	React.js / Next.js, Tailwind CSS
Backend	Node.js + Express.js, or Python + Flask/FastAPI

Database / Knowledge Base	MongoDB, MySQL, Firebase, SQLite, ChromaDB, FAISS, Pinecone, Supabase Vector, or JSON-based storage for a basic version
AI / LLM	OpenAI, Gemini, Claude, Groq, Ollama, or any other suitable model

Final Deliverables

- GitHub repository link
- Live demo link, if deployed
- README file with setup instructions
- List of data sources used
- Short architecture explanation
- Screenshots of the chatbot
- At least 10 sample questions and answers
- Demo video or brief walkthrough
- Mention of limitations and safety precautions

Evaluation Criteria

- Understanding of the problem statement
- Quality and reliability of healthcare data used
- Use of Generative AI
- Retrieval/RAG implementation
- Backend structure and API design
- Frontend interface and user experience
- Safety handling for healthcare-related queries
- Source/reference handling
- Code quality and folder structure
- Documentation and explanation of the work

Important Instructions

- Do not build a simple static FAQ bot.
- Do not hardcode answers.
- Do not rely only on generic AI responses.
- The bot must use the healthcare data collected by you.
- Do not push API keys or private credentials to GitHub.

• Mention all data sources clearly.

• The final project should be presentable and demo-ready.

Final Note: This assignment is not only about building a chatbot. It is about showing how you think, collect data, design a system, use AI responsibly, and present a practical working solution.